MANIPAL UNIVERSITY JAIPUR
(University under Section 2(f) of the UGC Act)

Department of Mathematics and Statistics
Blueprints for the end-term examination, April 2025
MAS2001- Statistical Methods and Probability
Topic-wise marking distributions

Topics	Weightage of Marks
Probability Theory and Random Variables: Probability (Only One Lecture), Random variables, Cumulative distribution functions, Discrete random variables, Continuous random variables, Independent random variables, Probability mass and density functions, Expectation of random variables, Independent random variables, Chebyshev's inequality.	**14**
Central limit theorem. Probability distribution: Binomial, Poisson, Uniform, Normal.	**17**
Exponential Distribution, Theory of Estimation: Introduction, Parameter and Statistic, Properties of good estimator: unbiasedness, Consistency, Efficiency, Sufficiency, Maximum Likelihood estimator Method of Moment Estimation.	**17**
Bayesian estimation, Interval Estimation for means (large & small samples) Tests of Statistical Hypothesis: Introduction, Standard error, Statistical hypotheses, Critical region, level of significance, Type I and Type II errors (only Definitions)	**12**
Test about one mean (t-test for single mean) Test about equality of two means (t-test for independent samples), Test about equality of two means (t-test for dependent samples), Test of variances (F test), Chi square test (independence of attribute), Analysis of Variance (one way).	**20**

Tentative format of the ETE question paper

Section	No. of Questions	Marks of questions	Marks for each section
A	5	2	5*2=10
B	6	5	6*5=30
C	2	10	2*10=20
D	2	10	2*10=20

Course Coordinator-MAS2001